



Project Title - Optimizing hyperspectral payload design for Stratospheric Earth Observation platforms to support disaster response and hazard monitoring

Host University - Massey University

Industrial Partner - Kea Aerospace

Academic Supervisor(s)

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Project Description

Current Earth Observation face fundamental trade-offs between spatial resolution, spectral resolution, and revisit frequency, which can limit their effectiveness for rapid and reliable post-disaster environmental assessment. There is limited quantitative understanding of how these trade-offs affect the detectability of disaster-induced surface changes under realistic atmospheric, illumination, and sensor constraints.

Stratospheric unmanned aerial platforms, such as the Kea Atmos developed by Kea Aerospace, offer a new observational platform between satellites and low-altitude aircrafts. Operating in the stratosphere, these platforms can provide persistent, and high-resolution data, potentially improving monitoring of natural hazards in Aotearoa-New Zealand.

This PhD research will develop a quantitative framework to evaluate how hyperspectral sensor design choices influence the detectability of land-surface change associated with natural hazards. The focus is on defining physically and observationally grounded limits of detection, need to design a full operational system in the future.

The project has two research objectives:

(1) To quantify how hyperspectral spectral resolution (including band placement, bandwidth, and wavelength selection across visible to shortwave infrared regions) affects the detectability of surface changes associated with key natural hazards. This objective will establish detection thresholds under realistic surface reflectance, atmospheric, and noise conditions, providing a physically based understanding of spectral sensitivity.

(2) To determine the minimum spatial resolution required to reliably detect and characterise hazard related surface changes using physics-based and/or simulation-driven approaches. This will involve generating synthetic images to experiment how spatial scale influences detectability across different hazard types and environmental conditions typical of Aotearoa-New Zealand.

The research will focus on flood-related surface disturbance (including landslides) and volcanic eruptions. Additional anthropogenic hazard types (e.g., oil spills) may be included as transferability cases. Both simulated and real airborne hyperspectral datasets will be used to evaluate detection performance across varying environmental conditions, atmospheric states, and sensor configurations. This research will deliver a generalised, quantitative framework that links hyperspectral sensor design to detectability of disaster-induced surface changes. The outcomes will define spectral and spatial resolution thresholds required for hazard monitoring, provide clear sensor design constraints for stratospheric Earth Observation platforms. Collectively, these results will directly inform the design of lightweight hyperspectral payloads suitable for stratospheric deployment, including future integration with Kea Atmos platforms.

The work supports improved disaster preparedness and response capability in Aotearoa-New Zealand, where rapid and reliable assessment of earthquake, volcanic, landslide, and flood events is critical for infrastructure resilience and emergency management. In doing so, it aligns with national priorities for strengthening environmental monitoring and hazard resilience, including the Aotearoa New Zealand Aerospace Strategy 2023-2030 and broader civil defence and emergency management objectives that emphasise improved Earth Observation capability for risk reduction and response. Through strong collaboration with Kea Aerospace, the project also ensures that scientific outcomes are directly translatable into future aerospace systems, supporting the development of next-generation stratospheric sensing capability consistent with Aotearoa-New Zealand's strategic investment in advanced aerospace technologies and sovereign Earth Observation infrastructure.

Student Time Split

University base for student (university, campus, department):

Massey University, Manawatu Campus, School of Agriculture and Environment

Industry base for student (company, site, address):

Kea Aerospace, 10 Sir William Pickering Drive, Christchurch 8053, New Zealand

Expected Time Split Between University and Industry Partner (in months):

24 months at Massey University and 12 Months at Kea Aerospace

Rationale for Time Split

The project has a planned split of 12 months based at Kea Aerospace in Christchurch and 24 months based at Massey University, Manawātū campus in Palmerston North.

The 12-month industry placement at Kea Aerospace will provide essential exposure for the student to learn about Kea Atmos platform and its flight operations, and design. This period will further ensure the research is aligned with industry needs for lightweight, high-performance hyperspectral payload design. Close collaboration with engineers and mission operators at Kea Aerospace will directly inform realistic system constraints and validation scenarios used in the research.

The remaining 24 months at Massey University will focus on the academic development and research, including hyperspectral imaging, physics-based modelling, and analysis of detectability thresholds under varying atmospheric, spectral, and spatial conditions. This phase provides the time and academic environment required for methodological development, experimentation, and synthesis across simulated and real datasets.

The proposed split ensures strong integration between applied aerospace development and fundamental research. Ongoing interaction with Kea Aerospace throughout the academic period will maintain alignment with evolving platform capabilities and ensure direct pathways for translation of research outputs into future stratospheric Earth Observation systems.

Application

To apply for this project please first read the guidance document and then complete the application form on the Applied Doctorates Scheme website.